



Deloitte.

DEVOPS ENGINEER



SCENARIO-BASED
INTERVIEW
QUESTIONS



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Scenario: Security vulnerabilities have been detected in production because no scans were performed during deployments. How would you integrate security into the pipeline?

- Key Points:
 - Add static application security testing (SAST) and dynamic application security testing (DAST) stages.
 - Use tools like SonarQube, OWASP ZAP, or Checkmarx.
 - Automate vulnerability scanning for container images.

Scenario: Your organization has set up a disaster recovery solution in the cloud but has never tested it. How would you test it without affecting production?

- Key Points:
 - Use a staging environment that mirrors production.
 - Simulate disasters like regional failures or data loss.

Scenario: Your relational database is struggling with write performance due to high concurrency. How would you solve this with sharding?

- Key Points:
 - Partition data logically (e.g., by customer ID or region).
 - Use database sharding tools or native sharded databases (e.g., Aurora, Cloud Spanner).

Scenario: Your API gateway needs to integrate with multiple authentication providers for different user groups. How would you design this?

- Key Points:
 - Use OAuth2 or OpenID Connect (OIDC) for authentication.
 - Implement identity federation with SAML or JWT tokens.
 - Cache auth tokens for repeated requests.

Scenario: Logs from multiple cloud providers need to be aggregated and analyzed in real time. How would you handle this?

- Key Points:
 - Use log aggregation tools like ELK Stack, Loki, or Cloud-native services (AWS CloudWatch Logs, Azure Log Analytics).
 - Normalize log formats for compatibility.

Scenario: Your e-commerce application experiences seasonal traffic spikes. How would you implement predictive scaling to handle this?

- Key Points:
 - Use historical traffic data to set scaling thresholds.
 - Automate predictive scaling with machine learning models.

Scenario: Your cloud costs are high, and an analysis reveals underutilized resources. How would you optimize costs without impacting functionality?

- Key Points:
 - Identify underutilized resources using cost management tools (e.g., AWS Cost Explorer).
 - Right-size instances and adopt instance scheduling.

Scenario: A security audit reveals vulnerabilities in your CI/CD pipelines, such as plaintext credentials and improper artifact handling. How would you address these?

- Key Points:
 - Store credentials in secure vaults (e.g., HashiCorp Vault, AWS Secrets Manager).
 - Sign and verify artifacts using GPG or similar tools.

Scenario: Your cloud and DevOps environments must comply with stringent regulatory standards. How would you ensure compliance?

- Key Points:
 - Use automated audit tools like AWS Config, Azure Policy, or GCP Cloud Security Command Center.
 - Create and enforce policies with tools like Open Policy Agent (OPA).

Scenario: Your application has multiple tiers (web, app, database) and needs strict network security between them. How would you configure this?

- Key Points:
 - Use security groups, network ACLs, or Azure NSGs to control traffic flow.
 - Segment networks with subnets for each tier.

Scenario: Your company operates on AWS, Azure, and GCP, and needs a centralized logging solution. How would you approach this?

- Key Points:
 - Use open-source tools like Fluentd or Logstash to aggregate logs.
 - Store logs in a cloud-agnostic data store (e.g., Elasticsearch, Datadog).

Scenario: A team uses Terraform for provisioning, but manual updates in the cloud environment create drift. How would you prevent and detect this drift?

- Key Points:
 - Use Terraform Cloud or terraform plan to regularly detect drift.
 - Implement a policy where all changes must go through code and CI/CD pipelines.

Scenario: Your infrastructure needs to comply with strict standards (e.g., PCI-DSS, ISO 27001). How would you enforce compliance in provisioning?

- Key Points:
 - Use policy-as-code tools (e.g., Open Policy Agent, AWS Config).
 - Automate checks in CI/CD pipelines for Terraform or CloudFormation templates.

Scenario: A stateful application in the cloud uses more resources than expected, increasing costs. How would you optimize this?

- Key Points:
 - Use managed services like RDS or Cosmos DB for databases.
 - Right-size compute and storage based on usage patterns.

Scenario: Your application is deployed across multiple regions, and resource over-provisioning is causing excessive costs. How would you optimize it?

- Key Points:
 - Analyze resource utilization using tools like AWS Cost Explorer or Azure Advisor.
 - Switch to autoscaling solutions for compute resources.

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